

DATABASES – LECTURE 3

Chapter II – Organizing data in external memory

- The evolution of organizing data
- Organizing data in files
- Models for data structuring in the database
- About data models
- Hierarchical data model
- Network data model
- Relational data model
- Object-oriented data model

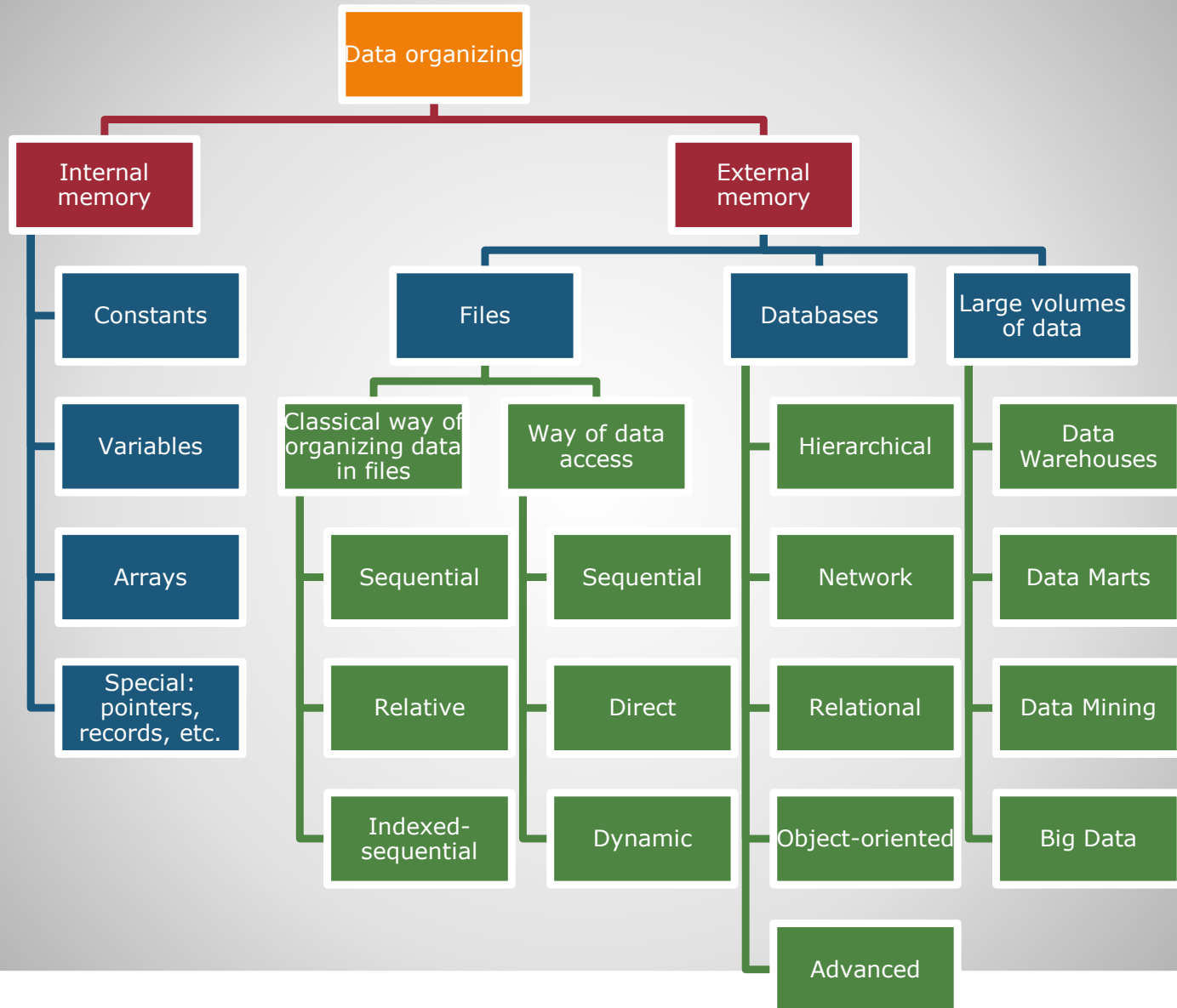
**Organizing data in external
memory**

- The evolution of organizing data

Organizing data = defining and structuring the data in the internal or external memory (in collections) and establishing links between data according to a data model.

Aspects Steps (1)	Way of organizing data (2)	The structure of the data (3)	Method of processing (4)	Redundancy (5)	Software (6)
1. Before 1965	Sequential files	Logical coincides with physical	Batch	Large, uncontrolled	Simple I/O operations (assembly and universal languages)
2 The '60s	Sequential, indexed, direct files	Logical and physical	Batch, on-line	Large, uncontrolled	Simple access keys (universal languages)
4. The '70s	Hierarchical, network databases	Logical, physical, conceptual	Batch, conventional	Decreases, controlled	Multiple key access, link between data, protection (DBMS)
3. Late '70s until now	Relational databases	Logical, physical, conceptual	Conventional, interactive	Small, controlled	Language retrieval, protection, concurrency (DBMS)
5. Late '80s until now	Object-oriented databases	Logical, physical, conceptual	Interactive	Minimum, controlled	OO programming languages (DBMS)

The evolution of organizing data



- Characteristics
- Structure
- Ways of organizing
 - Standard
 - Classic:
 - Sequential
 - Relative
 - Indexed-sequential
 - Special:
 - Partitions
 - Multi-indexed
 - Reversed
- Ways of data access
 - Sequential
 - Direct
 - Dynamic

Organizing data in files

- File = an organized collection of data that has in common quality, processing and purpose criteria.

1. Characteristics of a file:

- *Modifying* a file can mean: inserting, updating or deleting records;
- *The type of data* must be homogeneous in the file (it refers to the same entity in the real world);
- *Processing* the data file refers to the type and frequency of operations that are made on records;
- *The amount* of data in the files is measured in bytes.

2. The structure of a file:

- *the identifying part* = labels placed at the beginning and end of the file;
- *the data part* = homogeneous data collection belonging to the same real-world entity, structured as follows:

file → records → fields → values

3. Ways of organizing = the arrangement of records on physical support that involves data storage rules.

Categories of data organizing:

a) Standard

- It is the oldest and exists on any kind of computers.
- The record is made up of a series of characters arranged in a line supported by the peripheral.
- All programming languages recognize standard input and output files.

b) Classic (Elementary) - The organization is made on magnetic or optical environments.

- Types:
- SEQUENTIAL
 - ✓ The records are written in the file one after another without any predetermined order.
 - ✓ Locating a record is made by browsing all previous records (sequentially).
 - ✓ All operating systems and programming languages support sequential organization.
- RELATIVE
 - ✓ The records are arranged in a file one after the other and are numbered (by the system) from 0 or 1 (offset) to how many they are (execution number).
 - ✓ Locating a record can be made sequentially or directly through the execution number.

- INDEXED-SEQUENTIAL

- ✓ The records are arranged in the file in ascending order by a key (part of the record).
- ✓ The key is composed by one or more fields that uniquely identify a record.
- ✓ A table index is attached to the file. It links the key value to the physical address of the record.
- ✓ Locating a record is made sequentially and directly by the key:
 - the key of the requested record is compared with the indexes of the index table and the physical file that is containing the requested record is located directly;
 - in the located physical part, a sequential search of the required record is made.

c) Special (Complex)

- It is based on traditional (elementary) ways of organizing data.
- They are used in databases and file systems.
- Types (a few):
 - PARTITIONS
 - ✓ The file records are grouped into partitions that have a name.
 - ✓ In a partition the records are organized sequentially.
 - ✓ Are used in program libraries.
 - MULTI-INDEX
 - ✓ It is an extension of indexing organizing by using multiple keys chosen by the programmer.
 - ✓ The occupied space is larger.
 - ✓ It is used for files that require intense multi-retrievals.
 - REVERSED
 - ✓ Assume the existence of two files: basic and reversed.
 - ✓ The basic file contains the actual data and has sequential organization. It is the file in which the search is made.
 - ✓ The reversed file is built from the basic one (through inversion technique) and has relative organization. It is the file that helps the search.
 - ✓ The space required is about 3.5 times higher than for the basic file.

4. Ways of data access = ways to determine the place of a record in a file and depends on the organization mode.

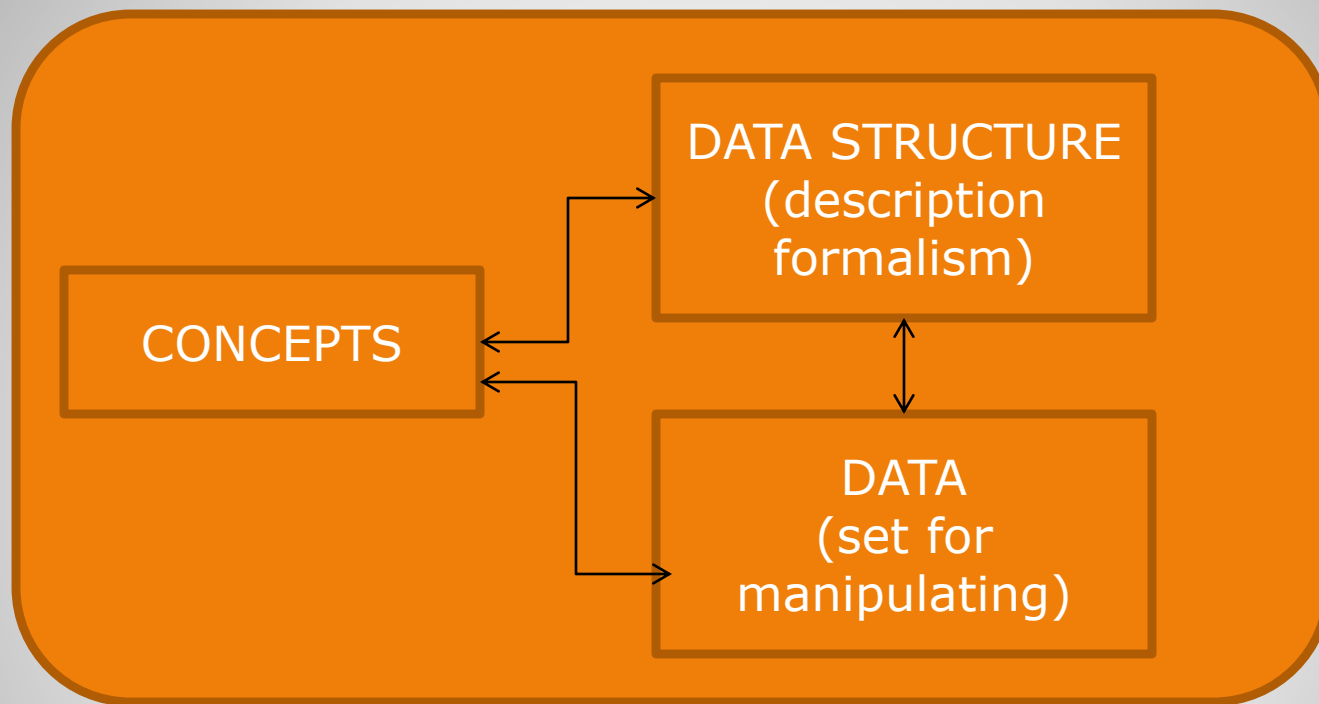
Types of access:

- SEQUENTIAL = location of records is made by going through all the records that preceded it.
- Sequential access is allowed for all file types.
- It is recommended for files which require in processing more than 50% of the total number of records.
- File ordering is recommended for optimization.
- DIRECT = location of a record is made by using a key that is defined by the programmer.
 - Direct access can be made by an execution number or by a key value.
- DYNAMIC = on one opening of the file, the records can be located alternately and repeatedly through sequential and direct access.

- **About data models**

- *The information*, that is represented in the computer's internal or external memory, can be structurally defined as follows:
- INFORMATION →(aims) ENTITY →(has as property) ATTRIBUTE (characteristic) →(has as measure) VALUE (domain)
- *The data model* = a set of concepts and tools for performing data collections' structure, according to which data will be represented.
- The data model is composed of:
 - concepts
 - a formalism for data description (data structure)
 - a data set for manipulation (data)
- *Schema [Conceptual schema]* = description of phenomena in the surrounding reality through entities and attributes (data type), together with all correlations (links) between them (constraints).
- DB schema is represented using a data model implemented by an appropriate DBMS.

Models for data structuring in the database



Data model

- **1. Defining the model's structure** (structural part):
 - defining entities and associated attributes;
 - defining relations (association) between entities.

The association may be of these types:

- one to one (1:1)
- one to many (1: M)
- many to many (M: N)
- explicitly undefined

Defining the data structure is made using a DDL of a DBMS.

- **2. Model's operators** (manipulation part) are acting on data for processing operations (merge, update etc.).

The operators are implemented using a DML in a DBMS.

- **3. Integrity rules** (coherence part) are constraints which are established in the description of data, which ensure data correctness and give the logic of the model.

The constraints are implemented with the DDL in a DBMS.

Elements (components) of any data model

Types of logical data models for DB:

- fundamental: hierarchical, network, relational, object-oriented;
- secondary (extended from the fundamental ones): distributed, multimedia, spatial, etc.

Types of data models for DB, considering the level they are used on:

- Conceptual model (in analysis phase):
 - describes the system in terms that can be easily understood by the users
 - it is used for initial specification of data and for interacting with the user
 - it is a way of establishing a connection between the developers and the users
- Logical model (in design phase):
 - specifies the DB structure (data collections and the links between them)
 - it is designed in such a way that it can be used for creating the DB
 - it is used for defining and maintaining the DB by the DBMS and for generating requests by the users
- Physical model (DBMS' task):
 - describes the way the logical data model will be represented when storing in the external memory (files, indexes, disks, etc.)
 - usually, it is automatically generated (by the DBMS) starting from the logical model

level 0

REC_TYPE 0

level 1

REC_TYPE 11

REC_TYPE 12

REC_TYPE 13

level 2

REC_TYPE 21

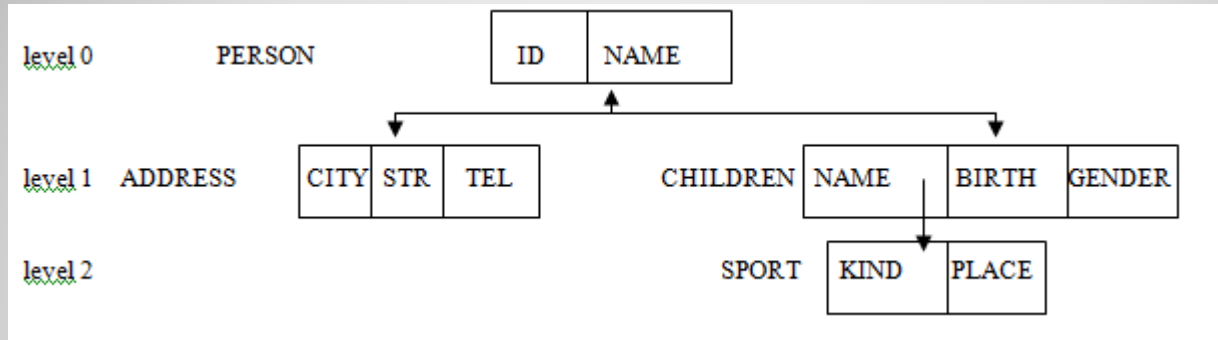
REC_TYPE 22

REC_TYPE 23

.....

Hierarchical model

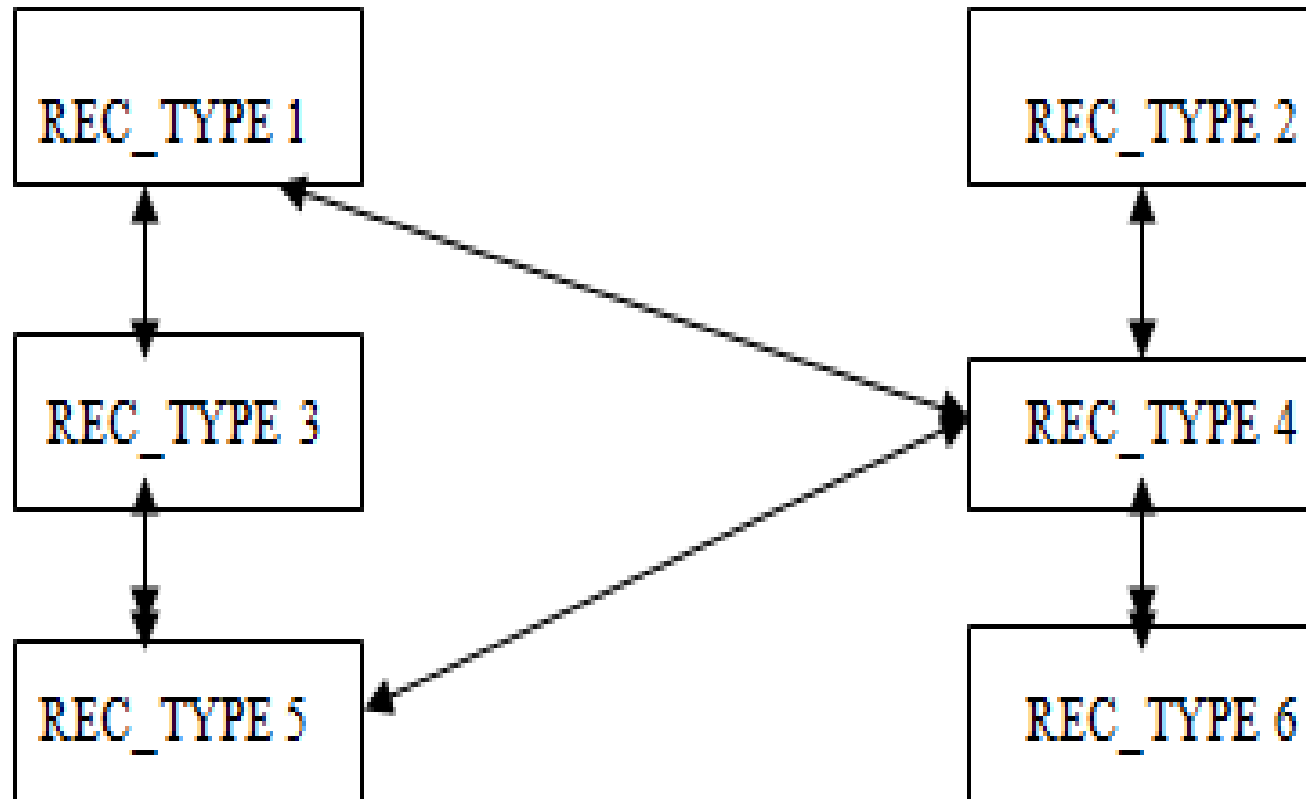
- the root record type PERSON with subordinate records type ADDRESS and CHILDREN. Record type CHILDREN has subordinate the record type SPORT.



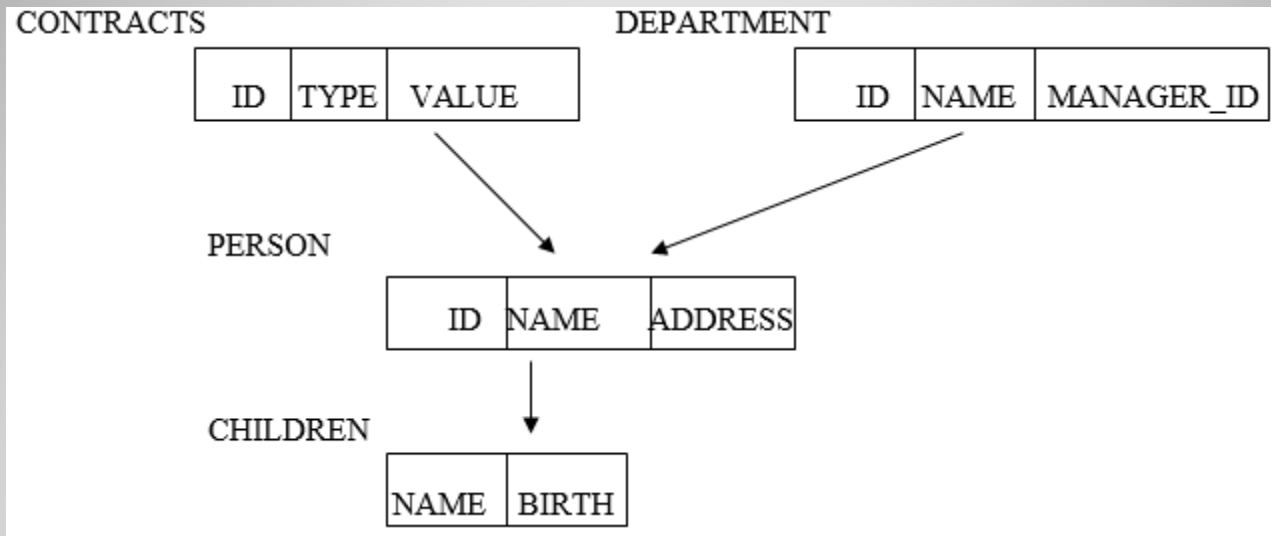
A hierarchical sequence of realizations for the above tree, may be:

PERSON	10	POPA MARIA	
ADDRESS	IAȘI	UNIRII	123456
CHILDREN	ANDA	220285	F
SPORT	SKI	MOUNTAIN	
	TENNIS	CITY	
CHILDREN	MARIA	221187	F
SPORT	ICE-SKATING	SKATE-PARK	

Example for creating a DB regarding the sport activities of people



Network model

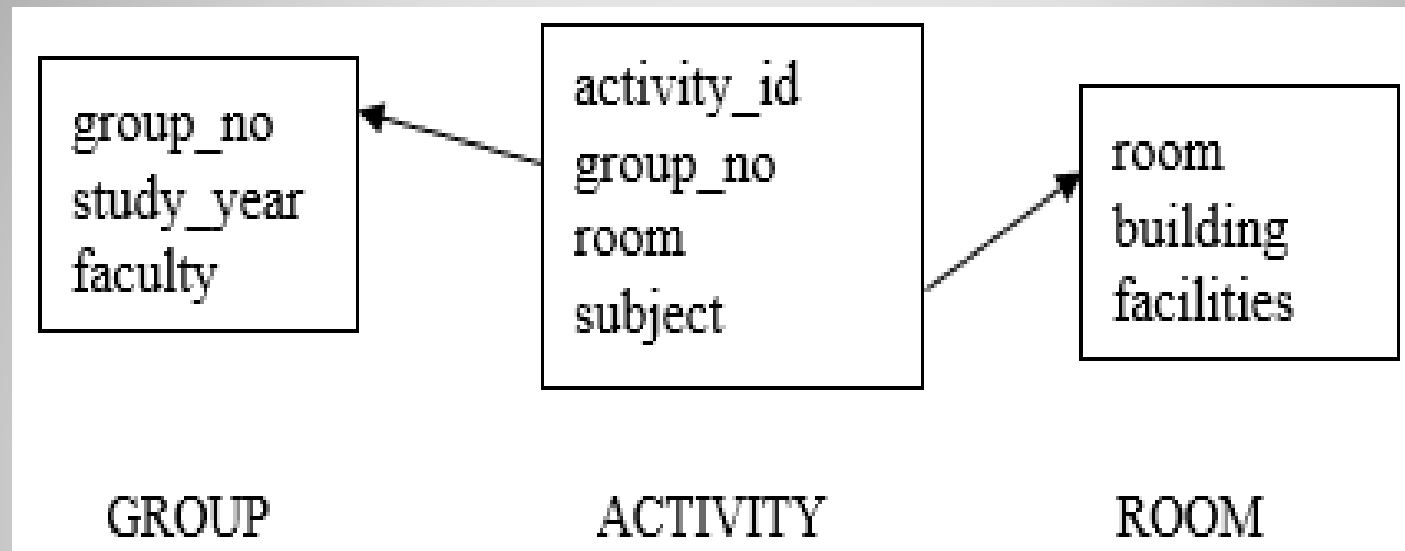


For this network structure a set type can be: the owner node DEPARTMENT and PERSON as a member node.

A realization of this set type can be:

DEPARTMENT	100	IT	001
PERSON	005	POPA ANDA	MOȘILOR, 22
PERSON	010	PETRE ION	EMINESCU, 10
.....			

Example: considering a staff DB with the entities (types of records): CONTRACT, DEPARTMENT, PERSON, CHILDREN.



Example of object diagram for the teaching activity, with classes of objects: teaching activities (ACTIVITY), group of students (GROUP), classroom (ROOM).

Example of object diagram

- Example of - hierarchical DBMS: **IMS** (Information Management System) developed by IBM.
- - network DBMS: **IDMS** (Integrated Database Management System) by CA Technologies, **Socrates** by Cincom Systems.
- - OODBMS: **O2** (O2 Technology), **Jasmine** (Fujitsu)

Relational data model

1. *Defining the structure of the relational model*

a) Defining the entities: the entities are represented by two-dimensional arrays called data tables or relations.

The concepts used are:

DOMAIN = a set of values, characterized by a name.

It can be explicit (lists the possible values D1: {M, F}) or implicitly (specifies properties of the values D1: {a/a $\in$ N}).

TABLE / RELATION = a subset of the Cartesian product of several domains, characterized by a name.

ATTRIBUTE = column of a table, characterized by a name. Takes values from a domain. Some can be keys.

TUPLE = line in a table and has no name and is numbered (starting from 1) by the system.

The order of lines (tuples) and columns (attributes) in a table shouldn't be important.

TABLE SCHEMA = table name followed by the list of attributes in brackets and for each attribute specifies the associated domain.

KEY = an attribute or set of attributes that serve to identify a tuple in a table.

Types of keys: primary / foreign, simple / composed.

RELATIONAL SCHEMA = table schema + keys + integrity constraints.

STUDENT

NAME: D1	YEAR: D2	BIRTH: D3
POPA A.	2	1992
PETRE M.	2	1992
POPESCU D.	3	1991

Example. Considering the STUDENT table with the attributes: NAME having the domain D1 (name of persons), YEAR having the domain D2 (years of study in faculty), BIRTH having the domain D3 (birth years).

Tuple: POPA A./2/1992

Table schema: STUDENT(NAME:D1,YEAR:D2,BIRTH:D3)

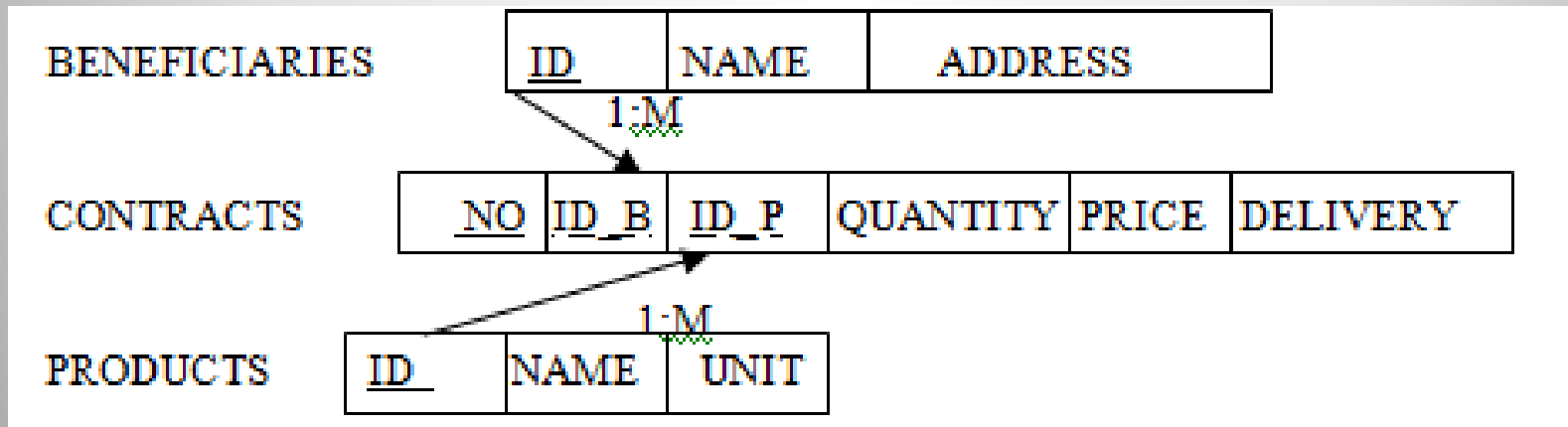
The number of domains is less than or equal to the number of attributes for a table (multiple attributes can take values from the same domain).

b) Defining the links between entities is made at logical level, by building associations between tables using connection attributes.

The links can be represented in the form of a structure chart called DB schema.

Example. Considering a DB about sales activity, for which the following tables can be identified: BENEFICIARIES, CONTRACTS, PRODUCTS. The links between tables are:

- a beneficiary may conclude several contracts (1: M);
- a product can be delivered through multiple contracts (1: M);
- a beneficiary can buy many products and a product can be delivered to multiple beneficiaries (M: N).



The M: N link is not displayed in the diagram. Key attributes are underlined.

Features of relational structure:

- The attributes involved in the links are either in associated tables, either in separate tables specifically designed for connections.
- The attribute in the original table is called foreign key and the attribute in the final table is called primary key.
- Possible connections are 1:1, 1: M, M: N.
- Potentially, any table can link to any table, by any attributes.
- The links are set when describing data (via DDL) using integrity constraints.

Practically, we can also establish dynamic links when executing a query.

Select prod_name from products, sales where prod_date=sale_date;

Products(id, prod_name,price, prod_date)

Sales(id, id_prod,sale_date,quantity)

2. *Operators* of relational model

a) **Relational algebra operators:**

- Standard: selection, projection, join, union, minus, Cartesian product, intersect, division;
- Extensions: complementary, splitting, etc.

Relational algebra operators act on tables in relational expressions. The operands are tables and the result is always a table.

- DIVISON

D1:{Alpha, Beta, Gamma, Delta}, D2:{1,2,3,4}

R table

A:D1	B:D2
Alpha	1
Alpha	2
Alpha	3
Beta	1
Beta	2
Gamma	1
Delta	1
Delta	3
Delta	4

R/r table

A:D1
Alpha
Beta

r table

B:D2
1
2

- COMPLEMENTARITY

D1:{alpha, beta}, D2:{1,2,3}

R table

A:D1	B:D2
alpha	1
alpha	2
beta	3

$\neg$ R table

A:D1	B:D2
alpha	3
beta	1
beta	2

- SPLITTING

D1:{alpha, beta}, D2:{1,2,3}, condition: $B > 2$

R table

A:D1	B:D2
alpha	1
alpha	2
beta	3

R1 table

A:D1	B:D2
beta	3

R2 table

A:D1	B:D2
alpha	1
alpha	2

b) Relational calculus operators:

- Can be tuple-oriented: connectives (conjunction $\wedge$, disjunction $\vee$, negation $\neg$), quantifiers (existential $\exists$, universal $\forall$);
- Or domain-oriented: idem as above.
- For relational calculus operators the operand can be a tuple or a domain.

3. *The integrity constraints* of the relational model.

- **a) Structural** = are defined by comparing the values in the tables:
- Unique key = in a table there should not be more tuples with the same value for the key assembly;
- Referential = in a table T1 that references a table T2, the foreign key values must be included in the values of primary key of T2 or be set to NULL (unspecified);
- Entity = in a table, the primary key attributes must not be set to NULL.
- The three above restrictions are minimal.
- In addition, there are other structural limitations that refer to data dependencies: functional, multi-value, join etc. (they are taken into account in the design techniques of RDB).

- **b) Semantic** = are defined by the data behavior and take into account the values from the database:
 - domain restriction = the domain corresponding to an attribute in a table must be between certain values; (CHECK)
 - temporary restrictions = the values of certain attributes are compared with some temporary values (results of calculations etc.).
- The semantic restrictions are quite general so they are handled either when describing data (e.g. CHECK clause), either when executing queries or other statements - outside the data model (e.g. IF statements).

The characteristics of the relational model:

- 1. It was proposed by IBM and revolutionized the representation of data in the database. It made the transition to a new generation (the second).
- 2. The model is simple, has a solid theoretical foundation being based on the set theory (assemblies) and mathematical logic.
- 3. All types of highly complex data structures from different fields can be represented using this model.
- 4. Model implementation is made at logical level by attributes with key roles.
- 5. The limitations of the relational model:
 - the great simplicity of the model makes it difficult to apply for new types of applications (multimedia, internet, etc.);
 - does not provide full logical independence of data to applications;
 - it cannot be used for applications of large volume and complexity;
 - can introduce excessive redundancy (when using the normalization design technique).
- 6. Relational database = a set of tables which store both the data and the links between it.
- 7. RDBMS: Oracle, IBM DB2, Microsoft SQLServer, MySQL, etc.